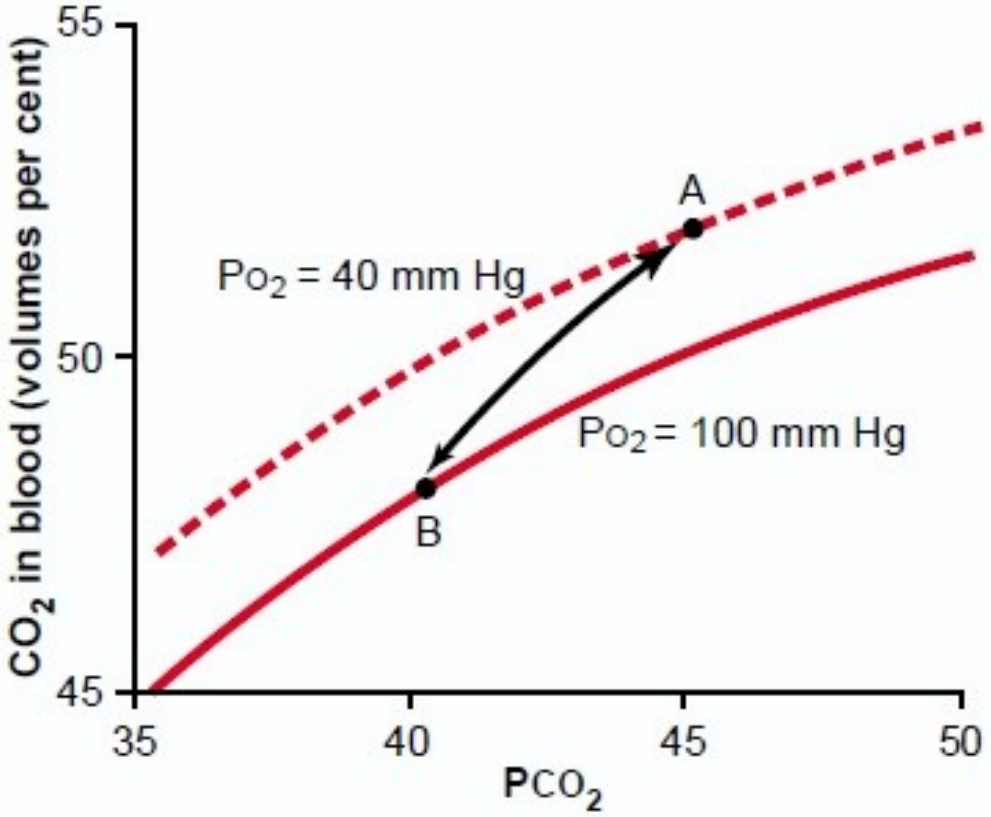
RESPIRATORY SYSTEM OSPE

DAY 4

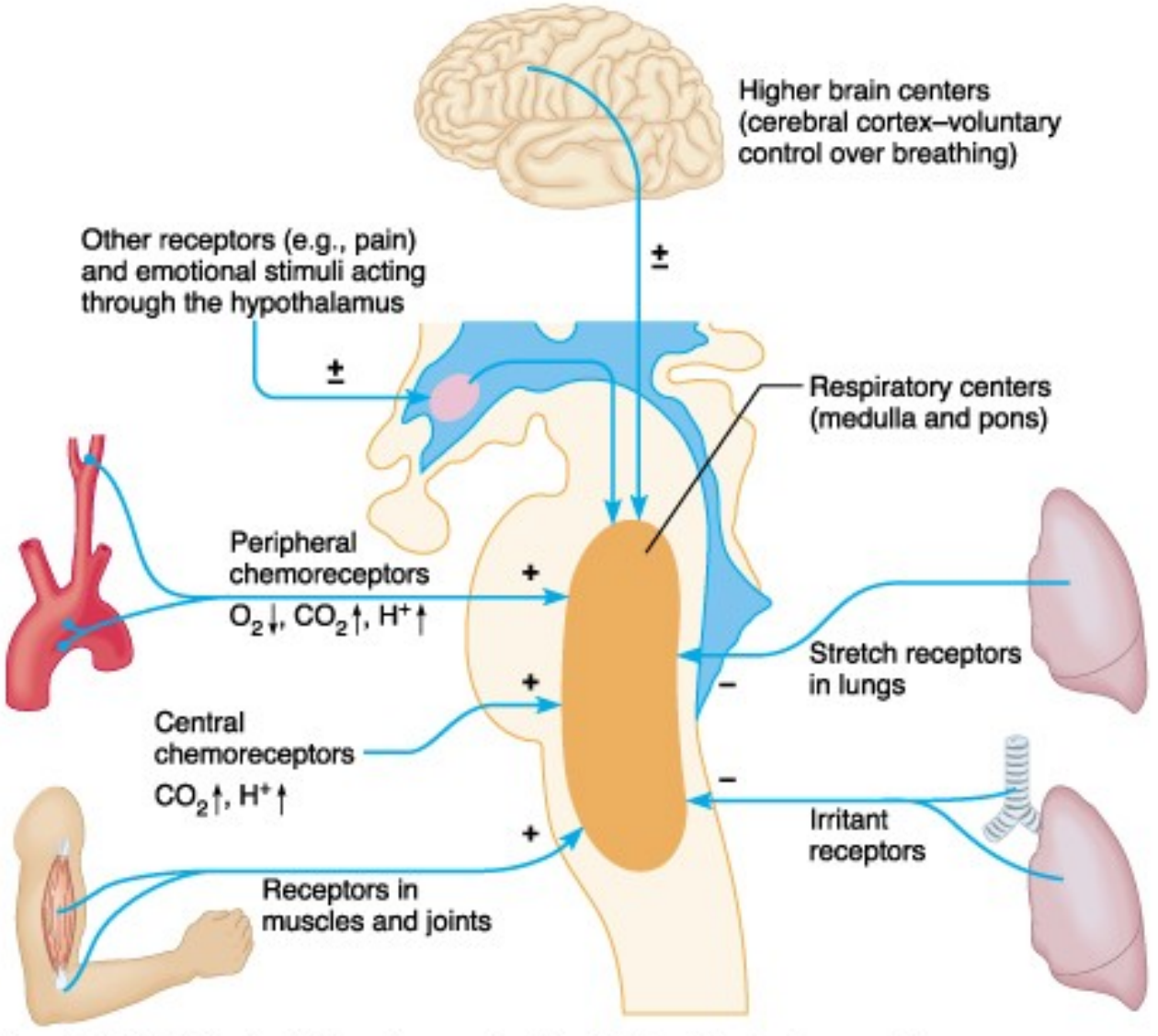
SPOT NO.1

1. Identify the physiological principle illustrated by the difference between the two curves.
2. In which site (lungs or tissues) does this effect enhance CO_2 unloading?
3. Explain briefly why deoxygenated blood carries more CO_2 at the same PCO_2.
4. State one mechanism by which CO_2 is transported in the blood.



1. Identify the primary stimulus detected by central chemoreceptors.
2. Name the major location of peripheral chemoreceptors.
3. Explain how an increase in arterial PCO_2 alters ventilation via central chemoreceptors.
4. State one condition in which peripheral chemoreceptors become the dominant drive for respiration.

SPOT NO.2



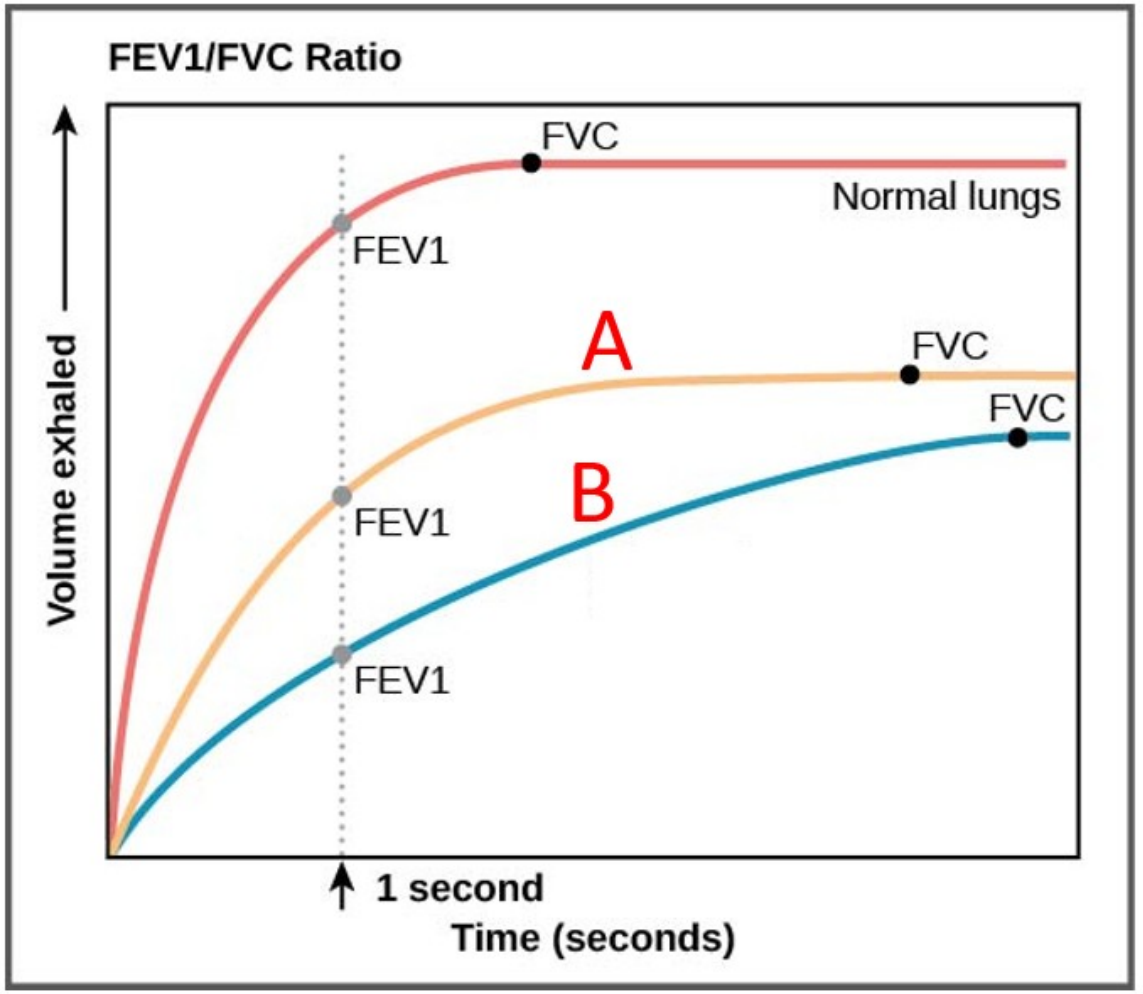
1. Identify the clinical sign shown in the picture.
2. What is the minimum concentration of deoxygenated hemoglobin required for this sign to appear?
3. Why may this sign be absent in severe anemia despite hypoxemia?
4. Name one condition in which this sign may occur despite normal arterial PO_2.

SPOT NO.3



SPOT NO.4

1. Identify the pulmonary function test represented in the graph.
2. Which curve (A or B) represents an obstructive lung disease?
3. How does the FEV_1/FVC ratio change in this condition?
4. Give one example of a disease showing this pattern.



SPOT NO.5

1. Identify the physiological phenomenon illustrated in the diagram.
2. Which gas becomes increasingly dissolved in body tissues during descent?
3. What happens if ascent to the surface is too rapid?
4. Name one condition or symptom that results from this event.

